

WEBSTER CITY, IA, USA

WMO: 725478

Lat: 42.436N

Lon: 93.869W

Elev: 342

StdP: 97.29

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.2	-26.2	0.4	-22.1	-23.0	0.5	-18.9	14.4	-8.5	12.8	-9.2	3.7	320	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.4	24.4	32.2	24.2	30.3	22.8	26.8	31.2	25.6	30.0	24.4	28.8	5.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	21.3	29.7	24.0	19.7	28.6	22.8	18.3	27.4	86.1	31.7	80.8	30.0	75.5	28.9	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.1	9.7	DB	-26.5	36.7	2.7	3.6	-28.4	39.3	-30.0	41.4	-31.5	43.5	-33.5	46.1	(4)
(5)			WB	-26.6	28.3	2.5	1.8	-28.4	29.6	-29.9	30.6	-31.3	31.7	-33.1	33.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-6.3	-4.8	2.2	9.5	16.2	21.8	23.5	21.9	18.0	10.8	3.0	-4.2	(6)	
	DBStd	11.94	6.78	7.02	6.76	5.50	4.93	3.55	3.26	3.13	4.67	5.46	5.94	6.53	(7)	
	HDD10.0	1973	505	416	254	73	7	0	0	0	2	56	221	439	(8)	
	HDD18.3	3758	763	648	499	267	101	9	2	6	63	241	461	697	(9)	
	CDD10.0	1747	0	1	14	59	199	354	420	368	241	81	9	1	(10)	
	CDD18.3	491	0	0	1	3	35	114	164	116	51	8	0	0	(11)	
	CDH23.3	4836	0	4	8	73	385	1120	1629	987	534	93	3	0	(12)	
	CDH26.7	1721	0	3	1	18	126	426	634	304	183	26	0	0	(13)	
(14) Wind	WSAvg	4.5	4.9	5.0	5.3	5.6	5.3	4.2	3.2	2.8	3.5	4.5	5.1	4.9	(14)	
(15) Precipitation	PrecAvg	960	27	27	55	94	126	155	131	125	71	60	47	31	(15)	
	PrecMax	1392	57	50	120	205	214	297	317	334	160	183	112	66	(16)	
	PrecMin	682	6	4	3	26	45	40	48	25	24	9	1	0	(17)	
	PrecStd	178	13	14	33	46	46	79	72	78	35	41	31	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	15.0	24.5	28.9	33.0	34.8	35.3	33.7	33.8	30.1	22.6	14.9	(19)	
		MCWB	7.0	10.5	15.5	17.8	21.7	24.4	26.4	25.8	22.6	19.8	15.0	11.4	(20)	
	2%	DB	7.1	10.0	19.1	25.2	30.0	32.7	33.7	32.0	31.2	26.1	17.9	9.9	(21)	
		MCWB	4.1	7.6	13.3	15.8	19.6	23.2	25.7	25.9	22.1	17.4	12.6	6.6	(22)	
	5%	DB	3.7	7.1	16.0	22.3	27.5	31.2	32.2	29.9	28.8	22.7	14.8	7.0	(23)	
		MCWB	1.9	4.7	11.7	13.9	18.3	22.5	25.2	24.0	21.1	16.2	9.9	4.7	(24)	
	10%	DB	2.1	3.8	12.3	19.1	25.1	29.0	30.2	27.9	27.0	20.0	12.1	3.9	(25)	
		MCWB	1.0	2.1	8.9	12.4	17.0	21.4	23.8	22.6	19.7	14.4	8.3	2.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	10.6	17.1	19.2	23.9	26.8	29.2	28.2	25.2	21.7	16.2	12.0	(27)	
		MCDB	11.7	13.9	22.8	25.7	30.0	32.4	33.4	32.4	30.6	28.1	21.1	14.4	(28)	
	2%	WB	4.1	7.3	14.3	17.0	21.6	25.4	27.5	26.1	23.7	19.3	13.3	7.0	(29)	
		MCDB	6.5	9.4	18.3	23.2	27.4	30.5	31.4	30.5	28.2	24.0	17.6	8.6	(30)	
	5%	WB	2.2	4.6	11.8	15.1	20.0	24.1	26.0	24.9	22.4	17.1	10.5	4.3	(31)	
		MCDB	3.7	6.8	15.7	20.5	25.4	29.1	30.3	28.7	26.6	21.2	13.9	6.5	(32)	
	10%	WB	0.8	2.2	8.6	13.2	18.6	22.7	24.9	23.6	20.9	15.0	8.1	2.3	(33)	
		MCDB	1.8	3.9	12.2	18.0	23.4	27.6	29.2	27.2	24.8	19.3	11.4	3.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.1	9.7	12.0	11.7	11.2	11.1	10.9	12.9	11.9	10.2	8.5	(35)	
		MCDBR	11.0	11.1	15.3	17.7	14.8	13.1	12.0	11.8	14.9	15.5	15.5	11.6	(36)	
	5% WB	MCWBR	8.9	8.7	9.9	9.7	7.0	6.2	6.6	7.5	7.7	8.7	10.1	9.4	(37)	
		MCWBR	11.1	10.3	14.2	14.7	12.6	11.5	10.7	10.8	12.0	12.8	14.4	11.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.294	0.320	0.374	0.403	0.422	0.432	0.416	0.387	0.341	0.297	0.278	(40)	
		taud	2.412	2.396	2.427	2.305	2.292	2.285	2.233	2.278	2.319	2.442	2.505	2.452	(41)	
	Ebn at Noon		864	913	928	893	872	854	840	842	841	846	846	841	(42)	
		Edh at Noon	77	92	102	124	129	131	136	126	113	88	70	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.88	2.90	3.78	4.74	5.60	6.21	6.38	5.45	4.47	2.99	2.02	1.52	(44)	
	RadStd		0.20	0.22	0.38	0.42	0.41	0.51	0.38	0.37	0.31	0.38	0.15	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page